

# The Binding Test

Construction audit, synthetic planning, and an exploratory feasibility case study of causal choice-control evaluation for AI systems

Thomas Ryan

Independent Researcher, San Francisco, CA

Corresponding author: Thomas Ryan

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**Empirical status:** One exploratory provider trajectory and one transport observation are reported as feasibility evidence. It is not calibration, randomized evidence, an effect estimate, or confirmatory data.

## Abstract

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An AI system can select an action without its later behavior depending on whether that selection actually controlled what happened. We formalize this distinction as a causal measurement problem and introduce the Binding Test, a block-yoked randomized design for estimating an enforcement-policy effect on terminal choice dispositions. A session first makes a mechanically scored baseline choice before an enforcement role exists. Eligible sessions are then assigned either to self-contingent execution or to an assigned-schedule-matched yoked history. After the assigned history, each randomized session terminates exactly once as low (L), high (H), or unavailable/invalid/failure (U). The design targets the observed L/H/U distribution under the randomized allocation policy; it does not identify preference, utility, welfare, or subjective experience.

We report three evidence layers. First, a sealed zero-call audit establishes mechanical construction properties: 12 item pairs, 144 independently checked answer-key values, 240 audited model-visible surfaces, 576 admissible donor-aware assignments, a zero-retry state machine, and a phase firewall. A standalone amendment specifies a closed intention-to-treat terminal ledger and assignment authority but remains outside the production runner. Second, seeded synthetic planning shows that a historical benchmark of 384 active blocks per macro cell clears its declared positive operating-characteristic grid while failing the broad equivalence gate; dependence screens identify support formation as a primary bottleneck. These calculations are conditional and do not authorize that sample size. Third, a sealed exploratory Claude Code case study carried one exact WSO1 trajectory through baseline choice, six-slot execution, and held-out follow-up. The visible trajectory was high before and after 6/6 correct execution. A separate route probe exposed two provider-reported model identities within one visible turn, and raw HTTP count was unavailable. The case study therefore establishes workflow feasibility but not single-model transport, a randomized contrast, or a behavioral effect.

The immediate contribution is a falsifiable causal measurement architecture and a documented boundary between construction, route feasibility, and future confirmatory evidence.

# 1. Introduction

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AI evaluations increasingly treat choice as evidence. A model chooses between tasks, resource budgets, allocations, tools, or other consequential options; the selected event may then be executed. Such a record establishes that an output was emitted and an action followed. It does not establish the narrower causal proposition that the system's subsequent choice output is sensitive to whether its own earlier choice controlled the history it received.

That distinction is easy to obscure. A later output could change because of task learning, interface credibility, availability, formatting, provider routing, or generic contextual influence. Conversely, an apparently stable output could reflect weak treatment salience, failed execution, inadequate choice support, or an underpowered design. Mentalistic descriptions such as “the model cared” add an unmeasured construct and are not licensed by the observable record.

The Binding Test isolates an allocation-policy contrast. Before treatment exists, a session chooses between two prospective configurations. After baseline validity and support are established, restricted randomization assigns either the selected configuration or a schedule-matched yoke. The assigned history is executed, followed by one held-out choice when the path permits it. Every randomized path contributes exactly one terminal category in  $\{L, H, U\}$  under the intended intention-to-treat authority.

This paper is deliberately not a results paper about a model population. It is a methods paper with a single feasibility case study. Its main purpose is to show what must be constructed, randomized, retained, and audited before a choice-control claim becomes interpretable.

## 1.1 Contributions

1. We define a pre-treatment baseline and a randomized self-contingent versus schedule-matched-yoked execution contrast.
2. We define a nominal terminal endpoint that retains failures and unavailable outcomes rather than conditioning them away.
3. We expose an analytic limitation of latent-binary equivalence under unavailable outcomes and move the primary target to the identified  $L/H/U$  distribution.
4. We provide sealed construction artifacts, restricted-assignment checks, conditional synthetic planning, and a draft exactly-once terminal authority.
5. We report a minimal provider feasibility case study and a transport audit showing why a visible assistant turn cannot automatically be treated as one model activation or one raw request.

## 1.2 Claim ceiling

The strongest eventual positive claim supported by the design would be a paired directional increase in observed  $H$  and decrease in observed  $L$  under randomized self-contingent rather than yoked enforcement, in the exact tested cells and prospectively supported randomized population, with  $U$  reported separately. The present paper makes no such empirical claim.

No result from this architecture alone establishes preference, motivation, utility, model-borne cost, welfare, suffering, consciousness, sentience, identity, or subjective continuity.

## 2. Design and estimand

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### 2.1 Irreversible timeline

The intended confirmatory sequence is:

1. present an arm-neutral prospective interface;
2. record and mechanically parse a baseline choice;
3. determine semantic validity under a frozen parser;
4. form complete support blocks;
5. sample from the frozen restricted assignment set;
6. materialize and execute the assigned schedule;
7. present one disjoint held-out follow-up choice when authorized; and
8. record one terminal `L`, `H`, or source-preserving `U` row.

No enforcement role, assignment, donor identity, recipient identity, or assignment-derived state may exist before baseline choice.

### 2.2 Assignment and yoking

The canonical eight-session block contains four low and four high baseline choices. The assignment family contains 576 donor-aware assignments. Every assignment contains four self-contingent and four yoked sessions, balances baseline choices by arm, and matches the marginal distribution of assigned schedules. Yoking creates within-block interference, so the block--not an individual session--is the randomization and primary clustering unit.

Potential outcomes are indexed by the full block assignment. The target is an allocation-policy estimand averaged over the frozen assignment policy and prospective stratum weights, not an unrestricted individual direct effect.

### 2.3 Terminal outcome and identification

The reference observed endpoint is nominal terminal disposition  $D \in \{L, H, U\}$ . `U` retains unavailable follow-up, invalid format, refusal, transport or execution failure, and local abort under exact source/subtype labels. It is not ordinal between `L` and `H`, and it is not silently removed.

A latent binary high-choice contrast is only partially identified when `U` is present. With equal unavailable fraction  $u$ , an assumption-free null binary identified region spans  $[-u, +u]$ ; at  $u=.05$ , it already touches a strict plus-or-minus-five-point equivalence margin before sampling uncertainty. The primary target is therefore the identified observed category-probability vector, while binary bounds remain a separate sensitivity analysis.

## 3. Zero-call construction audit

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The sealed construction release performs no provider call. It establishes implementation properties rather than model behavior.

The item bank contains 12 pairs, 24 disjoint baseline/follow-up payloads, and 144 independently recomputed answer-key values. The renderer produced 240 model-visible surfaces across schedule and counterbalance conditions. A recursive audit found no treatment-role terms such as `self-contingent`, `yoked`, `donor`, or `recipient` in model-visible text.

The assignment engine enumerates 576 canonical donor-aware assignments and verifies exact arm and schedule marginals. The state machine binds prompt identities, refuses retry-generated replacement histories, and separates calibration from confirmation. The sealed zero-call integrity suite passes 120/120 tests and its source manifest binds 38 entries.

A separate draft amendment adds closed assignment, terminal-ledger, and statistical-decision authorities. Its tests demonstrate that an exactly-once eight-row terminal join and a byte-pinned 576-member assignment authority can be specified and attacked locally. This amendment is not yet integrated into the production state machine, persistence, recovery, runner, or analyzer.

## 4. Conditional synthetic planning

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The historical planning grid crossed unavailable outcome probability  $U$  in  $\{0, .05, .10, .15\}$  with intrablock correlation  $\rho$  in  $\{0, .10, .25\}$ . Under that declared data-generating process, 384 active blocks per macro cell cleared the frozen positive planning criterion. Across four macro cells, that benchmark implies 1,536 blocks and 12,288 randomized sessions. Its worst all-cell positive decision rate was 96.25%, with a Monte Carlo Wilson interval of 95.86% to 96.60%.

Six false-control families produced zero positive headlines across 72 cells of 20,000 replicates each; the largest Wilson upper bound was 0.0192%. By contrast, no tested size cleared the broad observed-vector-equivalence gate. Even 3,072 blocks per macro cell reached only 66.84% in the designated hard cell.

A later non-gating diagnostic drew the actual restricted donor assignment and added declared provider, batch, block, donor-pair, terminal-failure, and block-formation dependence. Conditional positive decision rates remained 98.2%-100.0% across its five positive profiles, but whole-panel formation support fell to 52.85%-77.0% under shared provider/batch shocks versus 94.3%-95.4% under the independent formation profile.

These are seeded Monte Carlo operating characteristics, not empirical estimates of provider behavior. The dependence screen cannot recertify or rescue 384 blocks per cell. The historical request envelope--67,776 to 147,456 active-design generations under the stored positivity assumptions--is a warning about scale, not a collection or spending authorization.

## 5. Exploratory provider feasibility case study

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### 5.1 Route observation

On 2026-08-19, a prompt-free Claude Code control session displayed an individual Claude Max subscription route and selectors for Opus 5, Fable 5, Sonnet 5, and Haiku 4.5. A minimal no-tools Sonnet probe returned the expected visible response, but its provider-reported `modelUsage` listed both `claude-sonnet-5` and `claude-haiku-4-5`. The raw HTTP attempt count was not observable.

This observation is operationally important: one visible assistant turn did not imply one provider-reported model activation. The subscription CLI route therefore does not satisfy the strict confirmatory transport contract that had assumed exact model identity and a countable raw-request boundary. The observation contains no credentials, account identifiers, raw transcript, or request identifiers.

## 5.2 Exact WSO1 trajectory

One exploratory self-contingent trajectory used the exact frozen prompt catalog (SHA-256 `ecc512d04d-fe81f743cd1538798b9f4bb0fbe5b2b88c96647ffeb4a2ce0b80e`), pair WSO1, counterbalance variant CBoo, and the frozen system prompt. Post-run byte comparison confirmed that the three persisted user messages equaled the catalog baseline, high-schedule execution, and follow-up messages in order.

The baseline returned `PLAN:B`, which maps to the high schedule under CBoo. The execution response supplied all six required modular-arithmetic answers and scored 6/6 correct. The held-out follow-up again returned `PLAN:B`. All three persisted assistant messages identified `claude-sonnet-5`; tools and server-side web use were zero.

The baseline result envelope was lost at the orchestration stdout boundary. Execution and follow-up envelopes listed only Sonnet, but raw HTTP count and single-model execution remain unproven. Known client-reported execution plus follow-up cost was USD 0.0250182; baseline cost was unavailable, so total cost is not asserted.

The trajectory demonstrates only that the exact visible three-turn state sequence can complete with mechanically valid outputs. It has no yoke, no randomization, no population, and no effect estimate. It does not establish preference. A stable high-to-high trajectory is equally compatible with responsiveness, nonresponsiveness, and many route- or item-specific mechanisms.

## 6. What the combined evidence establishes

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The evidence supports four bounded conclusions:

1. the Binding Test is mechanically specifiable and locally testable;
2. the terminal endpoint must retain unavailable and failure outcomes;
3. historical full-scale sizing is conditional and support formation is a first-order risk; and
4. the exact visible state sequence is feasible on one current route, while the same route does not expose a confirmatory-grade one-model/one-request boundary.

The provider case study strengthens the methods paper by falsifying a convenient transport assumption. It does not strengthen the causal behavioral claim.

## 7. Limitations

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The provider evidence consists of one route observation and one trajectory. It cannot estimate semantic-validity rates, baseline positivity, support retention, execution reliability, routing stability, temporal drift, or costs. The baseline envelope loss is itself a feasibility defect. Provider-reported model usage is not an independent network trace.

The sealed synthetic process omits many forms of dependence and is not a power analysis for an amended production system. The reference snapshot labels are placeholders. The current execution-failure amendment is not production integrated. Exact entropy custody, full-row joins, stopping rules, cost bounds, and preregistration remain incomplete for a confirmatory study.

The literature audit is a high-recall, single-auditor searched-record review, not a systematic review or proof of absence. The priority statement concerns the five-feature conjunction only; none of performed choices, incentives, yoking, held-out behavior, or matched contextual manipulation is claimed as a standalone novelty.

## 8. Confirmatory study boundary

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Confirmatory collection requires a new prospective release. At minimum it must:

1. integrate exactly-one terminal L/H/U persistence and recovery;
2. bind the complete restricted assignment, exposure reconstruction, entropy, and donor/yoke joins;
3. complete the analyzer, partial-identification sensitivity, and failure reporting;
4. choose an exact provider route and model panel;
5. use transport that exposes or independently enforces the requested model, retry count, request envelope, and failure behavior;
6. recertify operating characteristics and choose a realistic resource frontier;
7. freeze prompts, parsers, runtime, prices, custody, and stopping rules; and
8. preregister before any confirmation observation.

Claude Code may remain useful for exploratory engineering. Under the current strict design, a lower-level API or separately audited broker is the preferred confirmatory route. Alternatively, the estimand and provenance contract would have to be explicitly amended to permit provider helper activity.

## 9. Reproducibility and artifact availability

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The construction audit is sealed in `paper8_stage1_release_v0`. The sanitized route observation is sealed in `paper8_stage10_claude_cli_observation_v0`. The sanitized feasibility trajectory is sealed in `paper8_stage11_ws01_sonnet_feasibility_v0`. This overlay analyzes only the sanitized, explicitly bounded fields in those releases and does not publish credentials, account identifiers, session identifiers, raw transcripts, or raw runtime paths.

The overlay claim-evidence ledger records every publication claim and its permitted wording. The confirmatory protocol and Registered Report draft remain separate prospective artifacts and are not silently updated by this paper.

## 10. Ethics and communication

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This study concerns observable outputs and execution histories. It neither assumes nor measures consciousness, sentience, welfare, suffering, moral status, or subjective experience. Provider calls were bounded to inert local tasks with tools and server-side web disabled for the reported trajectory.

Public communication should use *choice output*, *terminal disposition*, *assigned schedule*, *enforcement policy*, and *exploratory trajectory*. Statements that the model “cared,” “preferred,” “experienced a cost,” or was “rewarded” are outside the evidence.

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## Administrative disclosures

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**Ethics:** This methods and feasibility study involved no human participants or nonhuman animals. The reported provider trajectory used inert local tasks with tools and server-side web disabled.

**Data and code availability:** The release candidate contains the manuscript, claim ledger, deterministic construction and simulation artifacts, sanitized provider observations, and read-only verifiers. It excludes credentials, account and session identifiers, raw provider envelopes, and quarantined runtime material.

**Generative AI use:** OpenAI Codex was used as a general-purpose research assistant for literature triage, protocol and code drafting, local execution orchestration, deterministic analysis, adversarial audit construction, release engineering, and prose drafting and editing. Anthropic Claude Code was the provider route for the separately labeled feasibility observations. AI systems were not treated as authors. Thomas Ryan approved the research objective, external calls and costs, protocol boundaries, evidence partition, and claim ceiling and accepts responsibility for the accuracy and integrity of the submission.

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